DISEQUAZIONI IRRAZIONALI

n PARI

$$\sqrt[n]{f(x)} \geq g(x)$$

$$\begin{cases} f(x) \geq 0 \\ g(x) < 0 \end{cases} \quad \vee \quad \begin{cases} g(x) \geq 0 \\ f(x) \geq [g(x)]^n \end{cases}$$

$$\sqrt{x^2 - 1} > x + 2$$

① $\begin{cases} x^2 - 1 \geq 0 \\ x + 2 < 0 \end{cases}$ $\vee$ ② $\begin{cases} x + 2 \geq 0 \\ x^2 - 1 > (x+2)^2 \end{cases}$

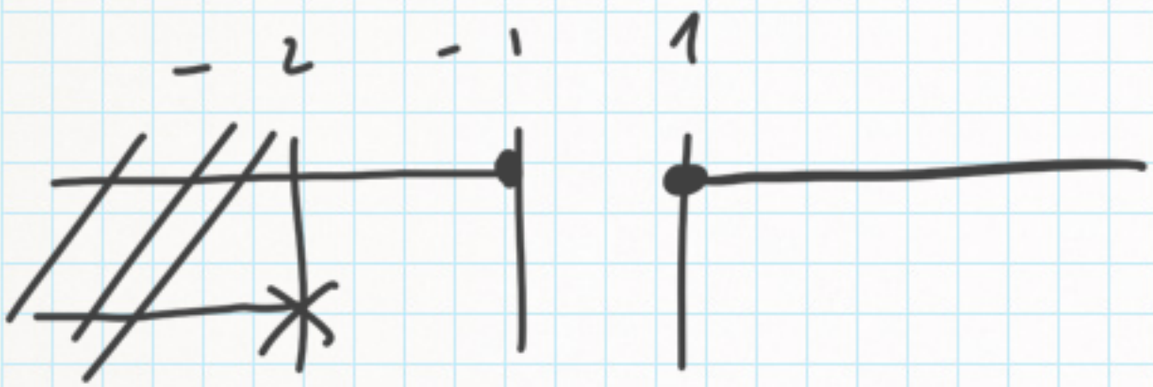
① $x^2 - 1 = 0 \iff x = \pm 1$

- $x^2 - 1 \geq 0 \iff x \leq -1 \vee x \geq 1$

- $x + 2 < 0 \iff x < -2$

$$\begin{cases} x \leq -1 \vee x \geq 1 \\ x < -2 \end{cases}$$

$$\boxed{x < -2}$$



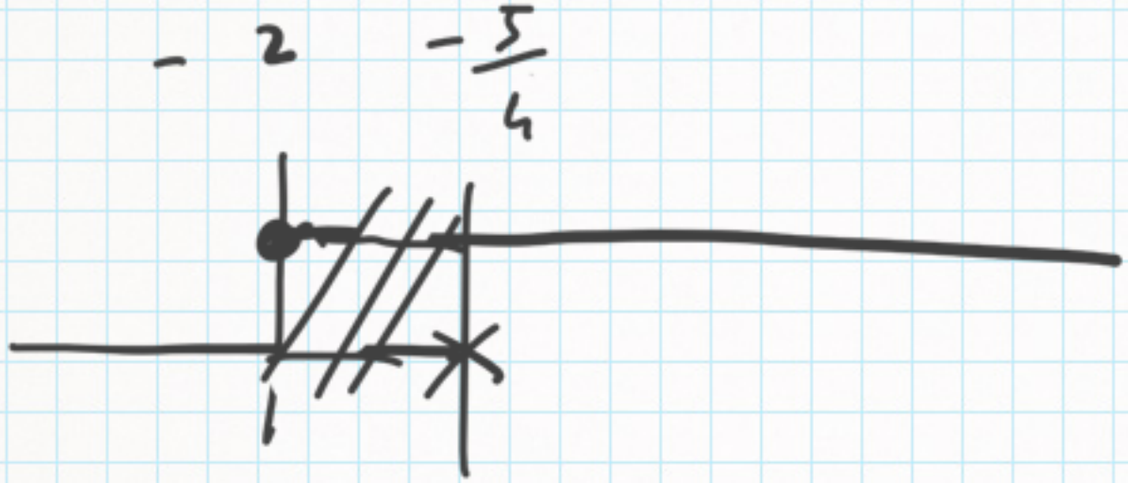
②

• $x+2 \geq 0 \iff x \geq -2$

• $x - 1 > (x+2)^2 = x^2 + 4x + 4$

$0 > 4x + 5$

$x < -\frac{5}{4}$



$$\begin{cases} x \geq -2 \\ x < -\frac{5}{4} \end{cases}$$

$$\boxed{-2 \leq x < -\frac{5}{4}}$$

① $]-\infty, -2[$

② $[-2, -\frac{5}{4}[$

① ∪ ② $=]-\infty, -\frac{5}{4}[$

Risolviamo $\sqrt[m]{f(x)} \leq g(x)$ m pari

$$\begin{cases} f(x) \geq 0 \\ g(x) \geq 0 \\ f(x) \leq [g(x)]^m \end{cases}$$

$$\sqrt{3x-1} < x+1$$

$$\begin{cases} 3x-1 \geq 0 \\ x+1 \geq 0 \\ 3x-1 < (x+1)^2 \end{cases}$$

$$\begin{cases} x \geq \frac{1}{3} \\ x \geq -1 \\ 3x-1 < x^2+2x+1 \end{cases}$$

$$\begin{cases} x \geq \frac{1}{3} \\ x \geq -1 \\ 0 < x^2-x+2 \end{cases}$$

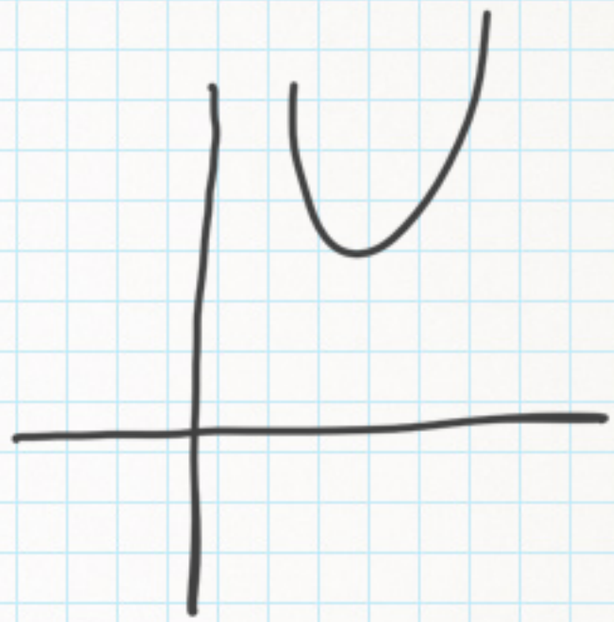
$$\begin{cases} x \geq \frac{1}{3} \\ x \geq -1 \\ x^2-x+2 > 0 \end{cases}$$

$x^2 - x + 2 = 0$

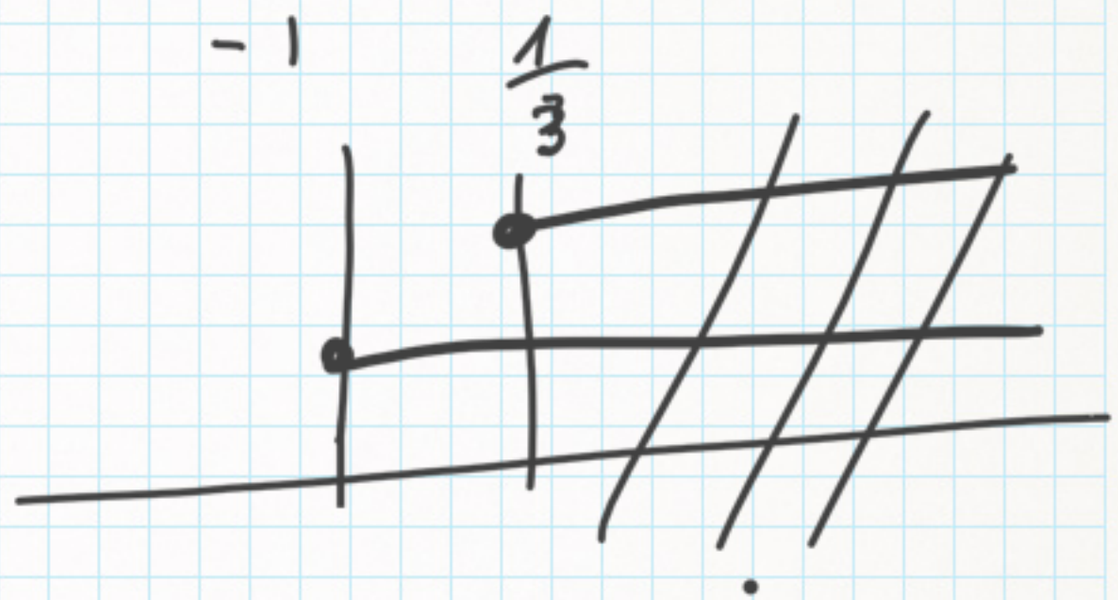
$\Delta = 1 - 8 = -7 < 0$

$x^2 - x + 2 > 0$

$\forall x \in \mathbb{R}$



$$\begin{cases} x \geq \frac{1}{3} \\ x \geq -1 \\ \forall x \in \mathbb{R} \end{cases}$$



$$\boxed{x \geq \frac{1}{3}}$$

• $x^2-3x+6 \geq 0 \iff x \leq 1 \vee x \geq 6$

$$x^2-4x-12 = 0$$

$$x = \frac{4 \pm \sqrt{64-48}}{2} = \frac{4 \pm \sqrt{16}}{2} = \frac{4 \pm 4}{2} \quad \begin{matrix} 4 \\ -2 \end{matrix}$$

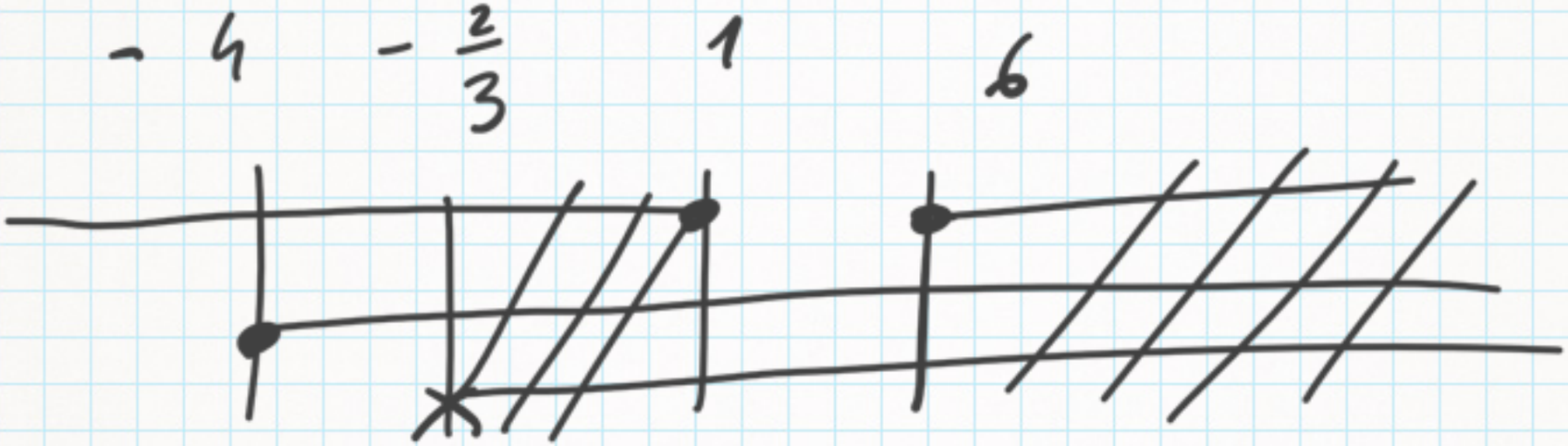
• $x+4 \geq 0 \iff x \geq -4$

$$\begin{cases} x^2-3x+6 \geq 0 \\ x+4 \geq 0 \\ x^2-3x+6 < (x+4)^2 \end{cases}$$

$$\sqrt{x^2-3x+6} < x+4$$

- $\cancel{x^2} - 7x + 6 < (x+4)^2 = \cancel{x^2} + 9x + 16$

$0 < 15x + 10 \iff x > -\frac{10}{15} = -\frac{2}{3}$



$-\frac{2}{3} < x \leq 1 \quad \vee \quad x \geq 6$

$$\frac{\sqrt{x+2} - 3x+1}{x+5} \leq 0$$

$$x\sqrt{x+2} - 3x+1 \geq 0 \quad \bullet$$ (RISOLVIAMO A PARTE)

$$x+5>0 \iff x>-5$$

$$\sqrt{x+2} - 3x+1 \geq 0$$

$$\sqrt{x+2} \geq 3x-1$$

① $\begin{cases} x+2 \geq 0 \\ 3x-1<0 \end{cases}$ $\vee$ ② $\begin{cases} 3x-1 \geq 0 \\ x+2 \geq (3x-1)^2 \end{cases}$

③ $\begin{cases} x \geq -2 \\ x < \frac{1}{3} \end{cases}$

$-2 < x < \frac{1}{3}$

$\begin{cases} x \geq \frac{1}{3} \\ x+2 \geq (3x-1)^2 = 9x^2 - 6x + 1 \end{cases}$

$0 \geq 9x^2 - 7x - 1$

ovvero $\quad 9x^2 - 7x - 1 \leq 0$

$$\left\{\begin{array}{l} x \geq \frac{1}{3} \\ 2x^2 - 7x - 1 = 0 \end{array}\right.$$

$$x_{1,2} = \frac{7 \pm \sqrt{49+24}}{12} = \frac{7 \pm \sqrt{73}}{12}$$

$$\frac{7-\sqrt{73}}{12} \leq x \leq \frac{7+\sqrt{73}}{12}$$

$$2x^2 - 7x - 1 \leq 0 \iff \frac{7-\sqrt{73}}{12} \leq x \leq \frac{7+\sqrt{73}}{12}$$

$$\frac{1}{3} \leq x \leq \frac{7+\sqrt{73}}{16}$$

$$\frac{7-\sqrt{73}}{12} \quad \frac{1}{3} \quad \frac{7+\sqrt{73}}{12}$$

① $-2 \leq x < \frac{1}{3}$

② $\frac{1}{3} \leq x \leq \frac{7+\sqrt{97}}{16}$

$\Rightarrow \quad -2 \leq x \leq \frac{7+\sqrt{97}}{16}$

↓

IL NUMERATORE È POSITIVO IN QUESTO INTERVALLO

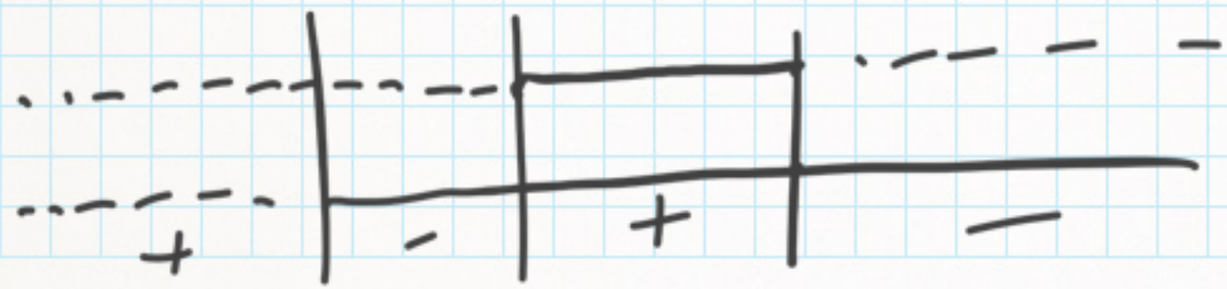
DEVO RISOLVERE

$$\frac{\sqrt{x+2} - 3x+1}{x+5} \leq 0 \quad (=)$$

$-5 < x \leq -2 \quad \vee$

$x \geq \frac{7+\sqrt{97}}{2}$

$-5 \quad -2 \quad \frac{7+\sqrt{97}}{16}$



- $\sqrt{27-3x} < \sqrt{25-x^2}$

$$\begin{cases} 27-3x \geq 0 \\ 25-x^2 \geq 0 \\ 27-3x < 25-x^2 \end{cases}$$

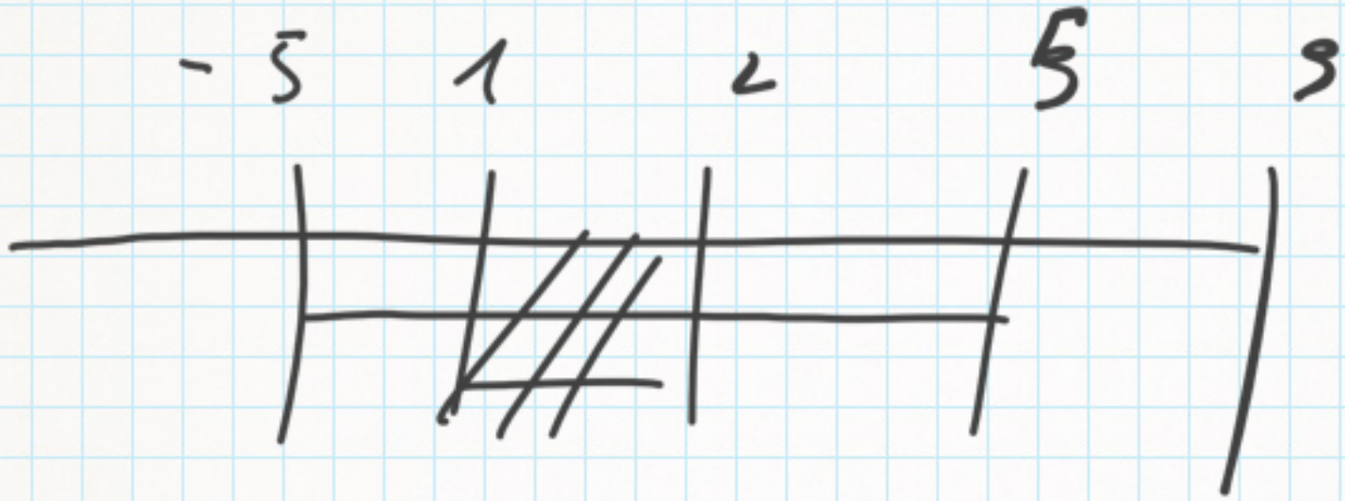
$$\begin{cases} -3x \geq -27 \\ x^2-25 \leq 0 \\ x^2-3x+2 < 0 \end{cases}$$

$$\begin{cases} x \leq 9 \\ -5 \leq x \leq 5 \end{cases}$$

$x^2-3x+2<0 \qquad 1<x<2$

$x^2-3x+2=0$

$x_{1,2} = \frac{3 \pm \sqrt{9-8}}{2} = \frac{3 \pm 1}{2} \begin{matrix} 1 \\ 2 \end{matrix}$



$$1 < x < 2$$

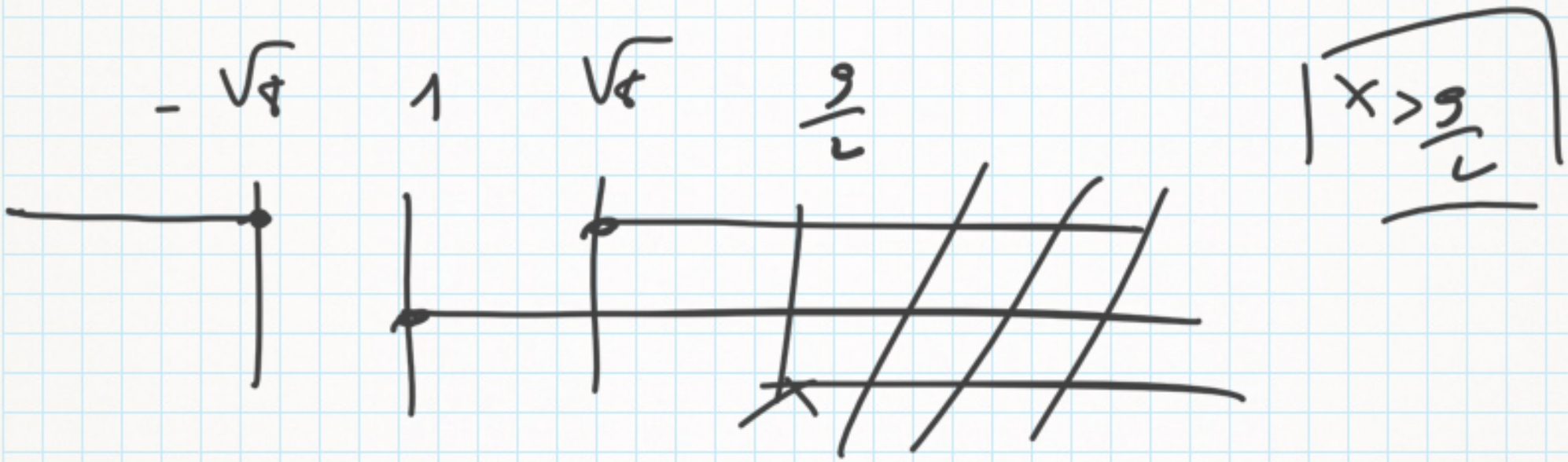
$$\sqrt[4]{x^2 - 8} > \sqrt{x - 1}$$

ELEVIAMO ALLA quarta

$$\begin{cases} x^2 - 8 \geq 0 \\ x - 1 \geq 0 \\ x^2 - 8 > \left((x-1)^{\frac{1}{2}} \right)^4 = (x-1)^2 \end{cases}$$

$$\begin{cases} x \le -\sqrt{8} \vee x \ge \sqrt{8} \\ x \ge 1 \\ x^2 - 8 > (x-1)^2 = x^2 - 2x + 1 \end{cases}$$

$$\begin{cases} x \le -\sqrt{8} \vee x \ge \sqrt{2} \\ x \ge 1 \\ 2x > 9 \Rightarrow x > \frac{9}{2} \end{cases}$$



$$\sqrt{x+2} - \sqrt{x-3} > 1$$

(SUFFICIENTE):

$$\sqrt{x-3} < \sqrt{x+2} - 1$$

$$\boxed{3 \leq x < 7}$$

.

Funzione esponenziale e funzione logaritmo

Sia $a > 0$, $a \neq 1$ (BASE)

Sia $q \in \mathbb{Q}$, $q = \frac{m}{n}$, $m \in \mathbb{Z}$, $n \in \mathbb{N}^*$

$$a^q = \left(a^m\right)^{\frac{1}{n}} = \left(a^{\frac{1}{n}}\right)^m$$

Sia $x \in \mathbb{R}$, $a > 1$

$$a^x := \sup\{ a^q \mid q \in \mathbb{Q},\ q \leq x \}$$

Se $0 < a < 1$ $\quad a^x := \dfrac{1}{\left(\frac{1}{a}\right)^x}$

Proprietà. Siano $a>0$, $a\neq 1$, $x\in\mathbb{R}$

(1) $a^x > 0$

(2) $a^0 = 1$

(3) Se $x<y$ allora $\begin{cases} a^x < a^y & \text{se } a>1 \\ a^x > a^y & \text{se } 0<a<1 \end{cases}$

(4) $a^x = a^y \iff x=y$

Possiamo quindi definire la funzione

ESPONENZIALE

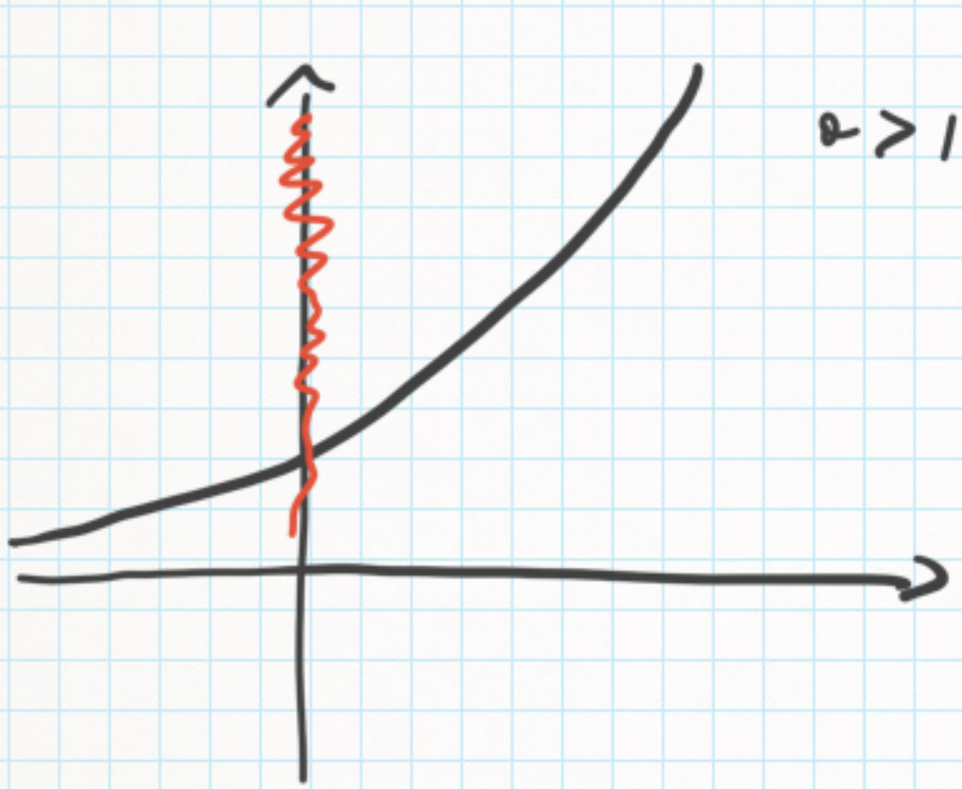
$a>0$, $a\neq 1$

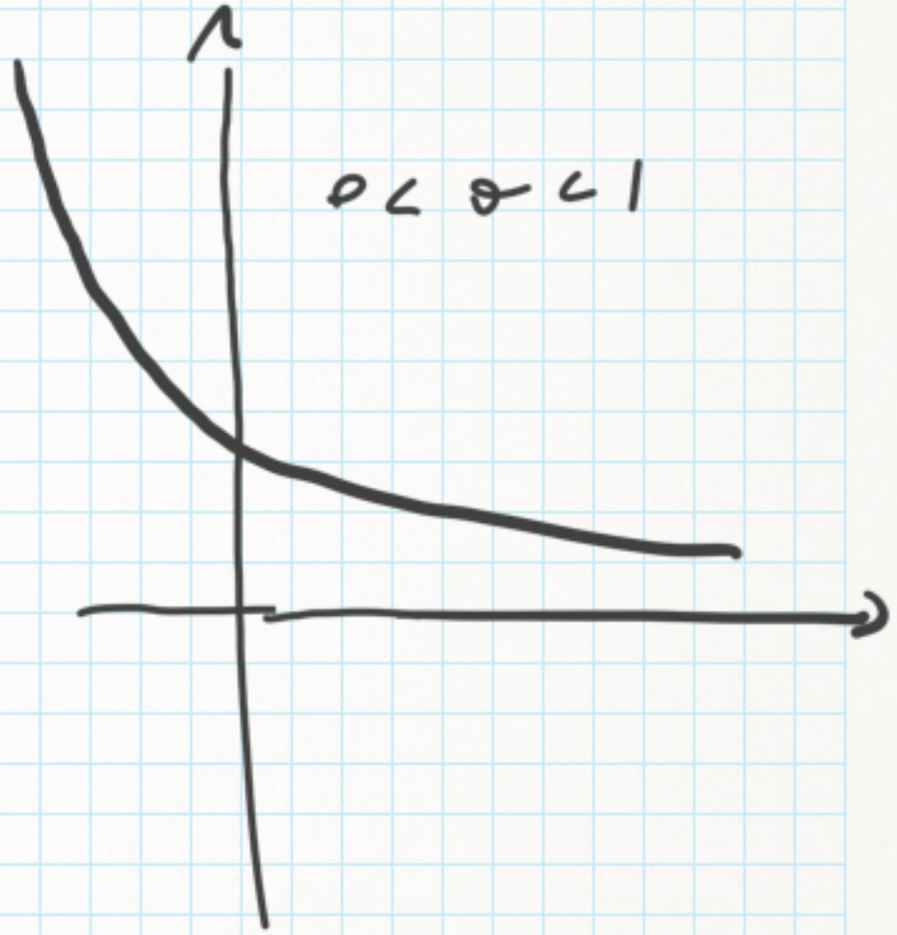
$$\exp_a : \mathbb{R} \to \mathbb{R}, \quad x \mapsto a^x$$

$$\exp_a(\mathbb{R}) =]0, +\infty[$$

- se $a > 1$ $\exp_a$ è STRETTAMENTE MONOTONA CRESCENTE

- se $0 < a < 1$ $\exp_a$ è STRETTAMENTE MONOTONA DECRESCENTE





$\exp_a$ è una funzione invertibile, perché è INIETTIVA (STRETTAMENTE MONOTONA)

La funzione inversa di $\exp_a$ è la funzione LOGARITMO IN BASE a

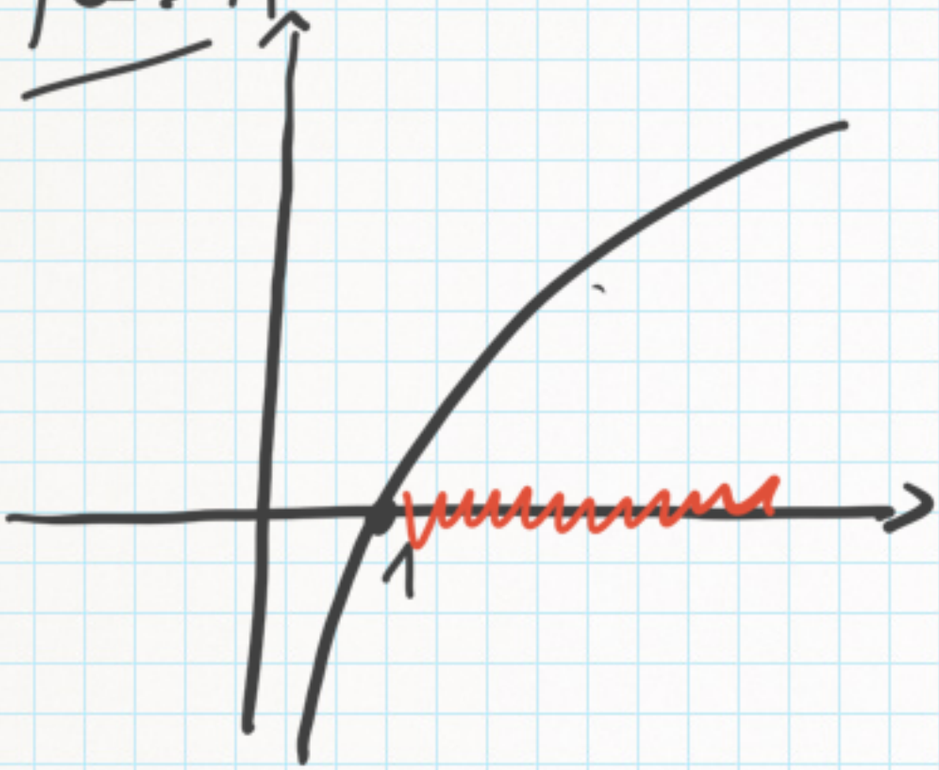
$$\log_a :]0, +\infty[\longrightarrow \mathbb{R}$$
$$x \mapsto \log_a x$$

ove per $\log_a x$ intendiamo l'unico valore $y \in \mathbb{R}$ t.c.

$$a^y = x$$

$$\left(a^{\log_a x} = x \qquad \log_a a^x = x \right)$$

$a > 1$

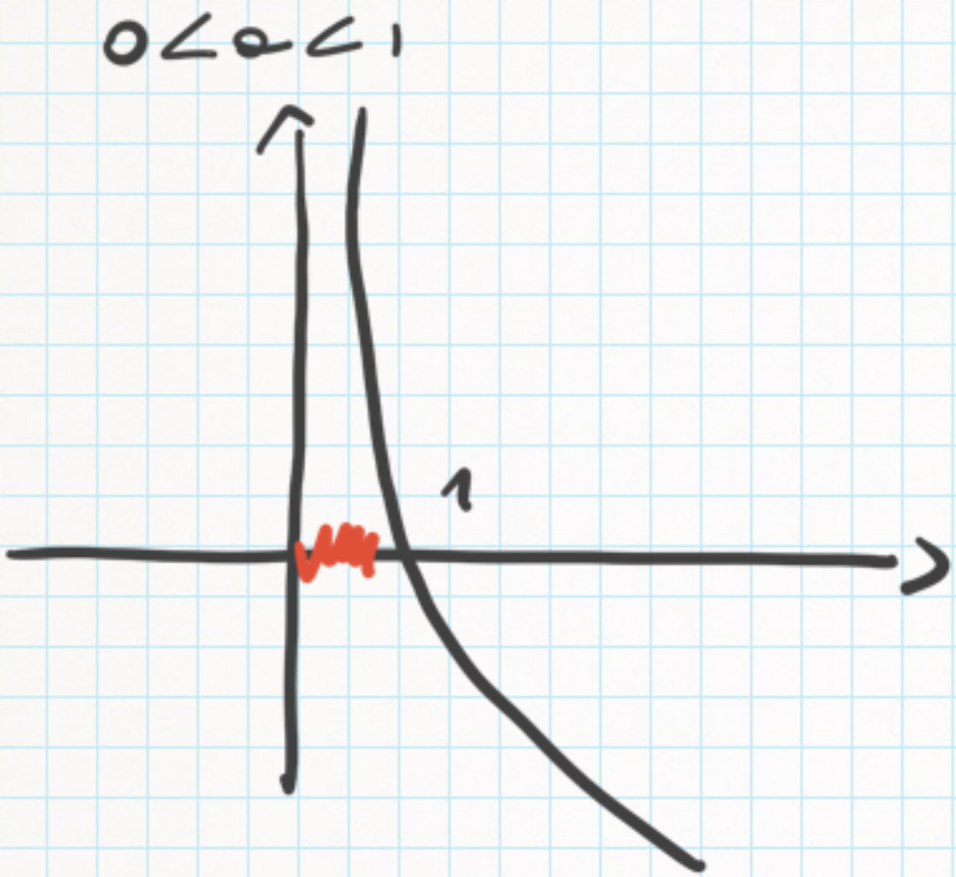


$\log_a 1 = 0$

$\log_a a = 1$

$\log_a x \geq 0 \iff x \geq 1 \qquad (a > 1)$

Se $a > 1$ $\log_a x$ è una funzione strettamente monotona crescente



$\log_a x \geq 0 \iff 0 < x \leq 1 \qquad (0 < a < 1)$

Se $0 < a < 1$ $\log_a$ è una funzione strettamente monotona decrescente

- $4^x = 16 = 4^2 \iff x = 2$ in quanto

in quanto la funzione $\exp_a$ è INIETTIVA

- $4^x = 20$ Applichiamo $\log_4$ ad entrambi i membri

$$\log_4 4^x = \log_4 20$$

$$x = \log_4 20$$

- $4^x < 20$

Applichiamo la fun $\log_4$ ad entrambi i membri

$$\log_4 4^x < \log_4 20$$

$$x < \log_4 20$$

La funzione $\log_4$ è STRETTAMENTE CRESCENTE, quindi il SEGNO DELLA DISUGUAGLIANZA RIMANE LO STESSO.

- $\left(\frac{1}{2}\right)^x > 32 = 2^5 = \left(\frac{1}{2}\right)^{-5}$

Ricapitolando

$$\left(\frac{1}{2}\right)^x > \left(\frac{1}{2}\right)^{-5}$$

Applichiamo $\log_{\frac{1}{2}}$ ad entrambi i membri.

RICORDANDO CHE $\log_{\frac{1}{2}}$ È STRETTAMENTE DECRESCENTE QUINDI BISOGNA INVERTIRE IL VERSO DELLA DISUGUAGLIANZA

$$\log_{\frac{1}{2}}\left(\frac{1}{2}\right)^{x} < \log_{\frac{1}{2}}\left(\frac{1}{2}\right)^{-5}$$

$$\boxed{x < -5}$$

$$5^{2x} - 5^{x+1} - 6 = 0$$

$$5^{2x} - 5 \cdot 5^{x} - 6 = 0$$

$$y = 5^x$$

$$y^2 - 5y - 6 = 0$$

$$y_{1/2} = \frac{5 \pm \sqrt{25+24}}{2} = \frac{5 \pm 7}{2} = \begin{cases} +6 \\ -1 \end{cases}$$

$$y^2 - 5y - 6 = 0 \iff y = +6 \vee y = -1$$

$$5^x = 6 \quad \vee \quad 5^x = -1 \rightarrow$$

$5^x > 0 \quad \forall x \in \mathbb{R}$

Risolviamo $5^x = 6$

$$x = \log_5 6$$

- $5^{2x} - 5^{x+1} - 6 < 0$

$y = 5^x$

$y^2 - 5y - 6 < 0$

$-1 < y < 6$

$-1 < 5^x < 6$

$$\begin{cases} 5^x > -1 \\ 5^x < 6 \end{cases}$$

$$\begin{cases} \forall x \in \mathbb{R} \\ x < \log_5 6 \end{cases}$$

$$\boxed{x < \log_5 6}$$

$$\cdot \left(\frac{1}{4}\right)^{x-1} + \left(\frac{1}{4}\right)^{x} - 5 > 0$$

$$\frac{\left(\frac{1}{4}\right)^{x}}{\frac{1}{4}} + \left(\frac{1}{4}\right)^{x} - 5 > 0$$

$$4\left(\frac{1}{4}\right)^{x} + \left(\frac{1}{4}\right)^{x} - 5 > 0$$

$$5\left(\frac{1}{4}\right)^{x} > 5$$

$$\left(\frac{1}{4}\right)^{x} > 1 \Rightarrow \log_{\frac{1}{4}}\left(\frac{1}{4}\right)^{x} < \log_{\frac{1}{4}} 1 = 0$$

$$\boxed{x < 0}$$

$$\log_{\frac{1}{3}} x \geq 2$$

$$\begin{cases} x > 0 \\ \left(\frac{1}{3}\right)^{\log_{\frac{1}{3}} x} \leq \left(\frac{1}{3}\right)^2 \end{cases}$$

APPLICHIAMO AD ENTRAMBI I MEMBRI

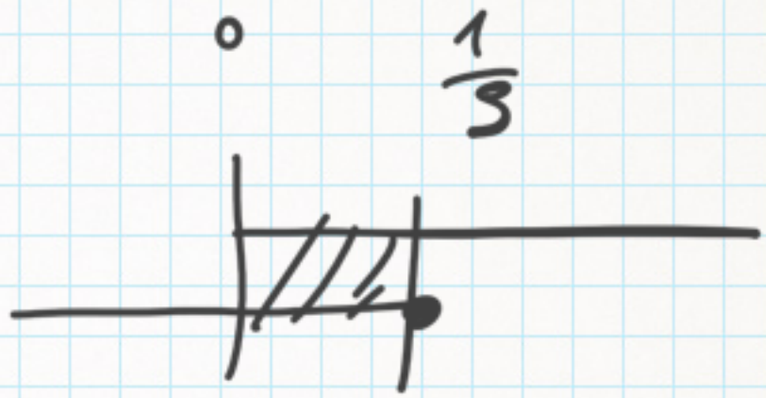
LA FUNZIONE $\exp_{\frac{1}{3}}$ MA DOBBIAMO

CAMBIARE IL VERSO DELLA DISUGUA-

GLIANZA PERCHÉ $\exp_{\frac{1}{3}}$ È

UNA FUNZIONE STRETTAMENTE DECRESCENTE

$$\begin{cases} x > 0 \\ x \leq \frac{1}{3} \end{cases}$$



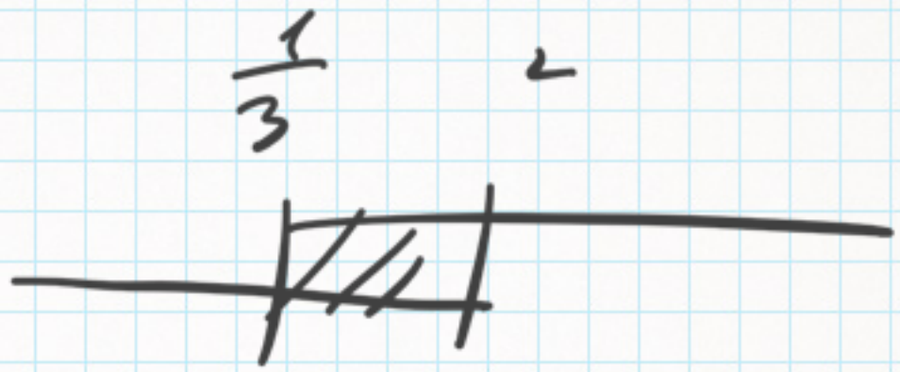
$$0 < x \leq \frac{1}{3}$$

- $\log_5 (3x-1) < 1 = \log_5 5$

$$\begin{cases} 3x-1 > 0 \\ \log_5 (3x-1) < \log_5 5 \end{cases} \qquad \begin{cases} 3x-1 > 0 \\ 3x-1 < 5 \end{cases}$$

LA DISUGUAGLIANZA NON CAMBIA VERSO PERCHÉ $a > 1$

$$\begin{cases} x > \frac{1}{3} \\ x < 2 \end{cases}$$



$$\frac{1}{3} < x < 2$$

$$\log_{\frac{3}{5}} (x^2 - x) < 1 = \log_{\frac{3}{5}} \frac{3}{5}$$

$$\begin{cases} x^2 - x > 0 \\ \left(\frac{3}{5}\right)^{\log_{\frac{3}{5}}(x^2-x)} > \frac{3}{5}^{\log_{\frac{3}{5}} \frac{3}{5}} \end{cases}$$

$$\downarrow$$

$$0 < \frac{3}{5} < 1$$

$$\begin{cases} \cancel{x^2 - x > 0} \\ x^2 - x > \frac{3}{4} \end{cases}$$ (in questo caso è superflua)

$$x^2 - x > \frac{3}{4}$$

$$4x^2 - 4x - 3 > 0 \iff x < -\frac{1}{2} \vee x > \frac{3}{2}$$

$$4x^2 - 4x - 3 = 0$$

$$x_{1,2} = \frac{4 \pm \sqrt{16 + 48}}{8} = \frac{4 \pm 8}{8} = \begin{cases} \frac{12}{8} = \frac{3}{2} \\ -\frac{4}{8} = -\frac{1}{2} \end{cases}$$

- $\log_3^2 x - 3\log_3 x > 0$

$\lceil \log_3^2 x = (\log_3 x)^2 \neq \log_3(x^2) = 2\log_3|x| \rfloor$

$\boxed{x>0}$

$y = \log_3 x$

$y^2 - 3y > 0 \iff y < 0 \vee y > 3$

$y^2 - 3y = 0 \iff y(y-3) = 0 \iff y = 0 \vee y = 3$

$\log_3 x < 0 \qquad \vee \log_3 x > 3$

$\begin{cases} x > 0 \\ 3^{\log_3 x} < 3^0 = 1 \end{cases} \qquad \vee \qquad \begin{cases} x > 0 \\ 3^{\log_3 x} > 3^3 = 27 \end{cases}$

$\begin{cases} x > 0 \\ x < 1 \end{cases} \qquad \vee \qquad \begin{cases} \cancel{x > 0} \\ x > 27 \end{cases}$

$0 < x < 1 \qquad \vee \quad x > 27$

DETERMINARE IL DOMINIO DELLA FUNZIONE

$$f(x) = \log_{\frac{1}{2}} \left(\frac{x^2 - x}{x+1} \right)$$

e determinarne la positività, ossia risolvere la disequazione

$$f(x) \geq 0.$$

DOMINIO: $\frac{x^2 - x}{x+1} > 0$

POSITIVITÀ: $0 < \frac{x^2 - x}{x+1} \leq 1$